

Maroua Arbouni

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OBJECTIVE

Seeking an IT/Software Engineering PFA internship to contribute to the digitalization, process automation, and optimization of information systems. As an AI & Digital Transformation Engineering student at ENSA Al Hoceima, I bring proven hands-on experience in full-stack web development, process automation, and enterprise architecture, and I am eager to deliver high-value, innovative IT solutions to drive operational excellence.

EDUCATION

National School of Applied Sciences (ENSA)

Engineering Cycle: AI & Digital Transformation

Al Hoceima, Morocco

2025 – Expected 2028

National School of Applied Sciences (ENSA)

Integrated Preparatory Classes

Al Hoceima, Morocco

2023 – 2025

PROJECTS

PFE Assignment & Management System | *Laravel, MySQL, Bootstrap, HTML, CSS, JavaScript*

- Engineered a full-stack web application to automate student-supervisor assignments under strict equitable constraints, parsing Excel uploads for 3 majors and all available professors, eliminating manual allocation errors.
- Built an automated defense scheduling module that dynamically coordinates supervisors, juries, students, rooms, and timeslots — reducing administrative overhead and preventing scheduling conflicts.

Building Permit Management Platform | *Laravel, MySQL, Python (NLP), Taiga*

- Designed and delivered a comprehensive web platform to fully digitalize building permit lifecycles, featuring online submission, multi-actor validation workflows, and digital archiving.
- Integrated Python-based NLP scripts to automatically extract applicant data and verify document compliance, optimizing municipal workload distribution and reducing processing time.

Universal Data Analyzer | *Python, Pandas, Streamlit, PySide6, SQLite*

- Engineered a Python application that consolidates multiple Excel uploads into unified summary files with dynamic chart generation — automating cross-file reporting for rapid business insights.
- Delivered a desktop GUI (PySide6) and a web dashboard (Streamlit) offering dual-interface access to analytical data.

Boarding School Services Digitalization | *ArchiMate, Zachman Framework, SWOT, PESTEL, Porter*

- Designed the full enterprise architecture (AS-IS to TO-BE) for a boarding school management platform covering student accommodation, secure payments, maintenance ticketing, and BI reporting, using ArchiMate and the Zachman Framework.

Web Platform “CozyWatch” | *Laravel, PHP, Bootstrap, HTML, CSS* | [GitHub](#)

- Developed a movie management platform using Laravel MVC, featuring full CRUD operations, user authentication, and a robust relational database schema.

Pharmacy Management System (MediControl) | *C, Linked Lists, Advanced Algorithms* | [GitHub](#)

- Developed an application in C utilizing linked lists for pharmacy stock management, order tracking, and automated low-stock notifications.

Custom Automata Compiler | *C, Flex, Bison, Graphviz*

- Created “Language A”, a custom language compiled with Flex and Bison, designed to generate and simulate automata with semantic error detection and Graphviz exports.

TECHNICAL SKILLS

Languages: PHP, Python, Java (OOP), C, JavaScript, HTML, CSS

Frameworks & Tools: Laravel, Bootstrap, Streamlit, PySide6, Git, Taiga, Linux

Databases & Systems: MySQL, SQLite, Information Systems & ERP Concepts

Architecture & Modeling: ArchiMate, Zachman Framework, SWOT, PESTEL, Porter, UML, Merise

Industry 4.0: openPLC (Digital Industry), Process Automation, NLP Integration

Languages Spoken: Arabic (Native), French (Fluent), English (Fluent), Korean (B2)